



Apache Spark at-a-glance

빅데이터분석
천세진

목표

- Spark shell 시작하기
- ML 알고리즘 사용하기
- HDFS로부터 데이터셋 탐색하기
- Spark SQL, Spark Streaming



Chapter

- Spark 시작하기
- Spark History
- Spark Essentials
- Spark Examples

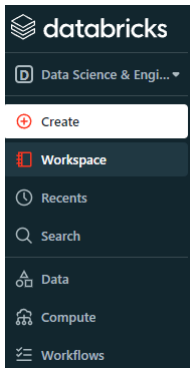


Databrick 맛보기

01 시작하기

Databricks 접속

■ Cluster에 따라 Notebook 생성



RDD 생성해보기

■ 데이터 생성하기

```
val data = 1 to 10000
```



RDD 생성해보기

■ 데이터 생성하기

```
val data = 1 to 10000
```

■ RDD 기반 데이터를 생성하기

```
val distData = sc.parallelize(data)
```

■ 10보다 이하 값에 대해서 필터 선택하기

```
distData.filter(_ < 10).collect()
```



Cluster 정보 확인

helloworld

Python

apache s park
 File
Edit
View: Standard

Attached cluster:

apache s park
 [Detach](#)
[Restart Cluster](#)
[Detach & Re-attach](#)
[Spark UI](#)
[Driver logs](#)
[Terminal](#)

15.25 GB · 2 Cores · DBR 9.1 LTS · Spark 3.1.2 · Scala 2.12



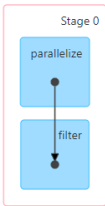
Spark UI 확인

Jobs Stages Storage Environment Executors SQL JDBC/ODBC Server Structured Streaming

Details for Job 0

Status: SUCCEEDED
Submitted: 2022/11/04 13:05:01
Duration: 2 s
Job Group: 1535292210689047424_4907544980669342435_9efa9a647f654862aadbad38e6ea57c
Completed Stages: 1

► Event Timeline
 ▼ DAG Visualization



▼ Completed Stages (1)

Page: 1

Stage Id ▾	Pool Name	Description	Submit
0	1535292210689047424	distData.filter(< 10).collect() collect at command-1867229373142010:1	2022/1 + details



Driver logs

Spark Driver Logs

Spark driver logs — Recent log files

All

☒ Auto-fetch data

Log file	Log type	Size
stdout	Standard output	380.96 KB
stderr	Standard error	2.63 KB
log4j-active.log	Log4j output	289.49 KB





Spark Deconstructed

01 시작하기

Log file

```
1  ERROR   php: dying for unknown reasons
2  WARN    dave, are you angry at me?
3  ERROR   did mysql just barf?
4  WARN    xylons approaching
5  ERROR   mysql cluster: replace with spark cluster
```



Upload data

Upload Data

Uploaded Files

Spark API Format

File API Format



dbfs:/FileStore/shared_uploads/sjchun@dau.ac.kr/log.txt

Copy

Done



Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

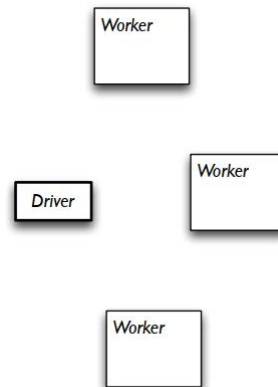
// action 2
messages.filter(_.contains("php")).count()
```



Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()
```



Log Mining Example

■ RDD operator graph를 확인 가능

```
scala> messages.toDebugString
res5: String =
MappedRDD[4] at map at <console>:16 (3 partitions)
  MappedRDD[3] at map at <console>:16 (3 partitions)
    FilteredRDD[2] at filter at <console>:14 (3 partitions)
      MappedRDD[1] at textFile at <console>:12 (3 partitions)
        HadoopRDD[0] at textFile at <console>:12 (3 partitions)
```

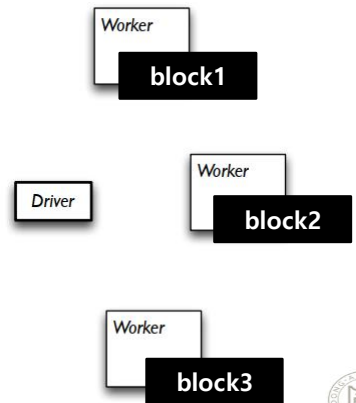


Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()
```

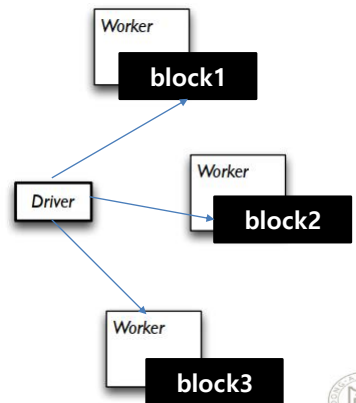


Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()
```

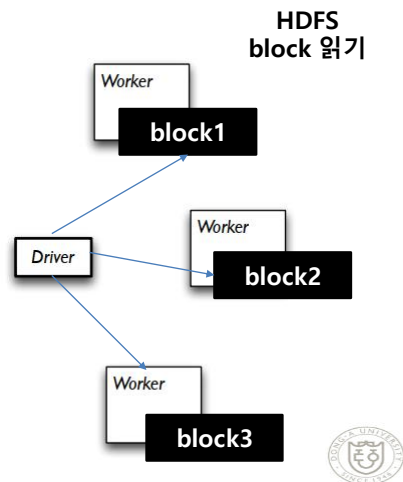


Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()
```

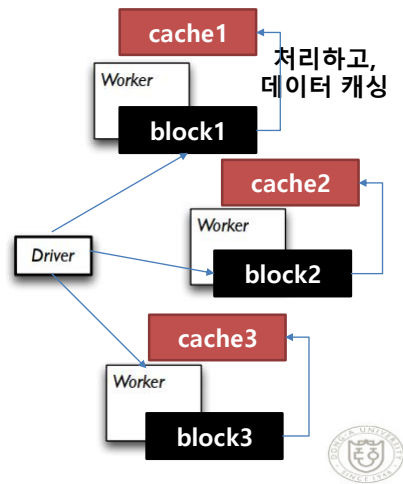


Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()
```

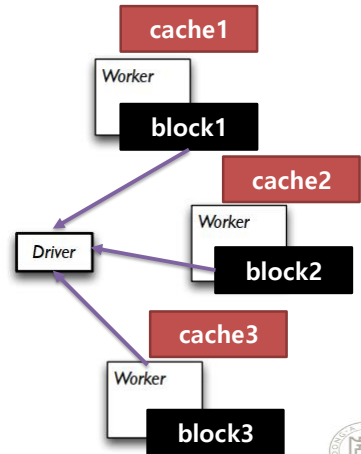


Log Mining Example

```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()
```



Log Mining Example

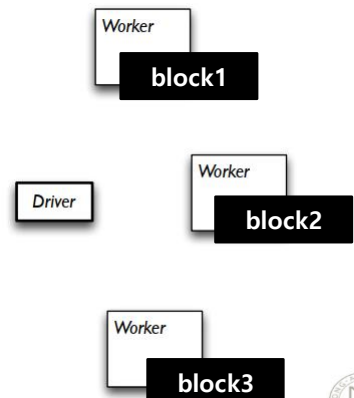
```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

// action 2
messages.filter(_.contains("php")).count()
```

실행 완료



Log Mining Example

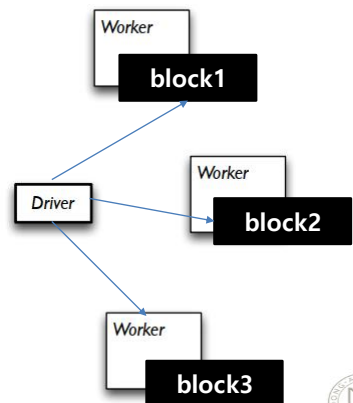
```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.filter(_.contains("mysql")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

// action 2
messages.filter(_.contains("php")).count()
```

실행 완료



Log Mining Example

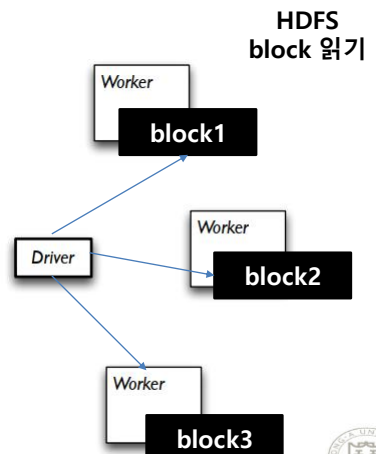
```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.filter(_.contains("mysql")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

// action 2
messages.filter(_.contains("php")).count()
```

실행 완료



Log Mining Example

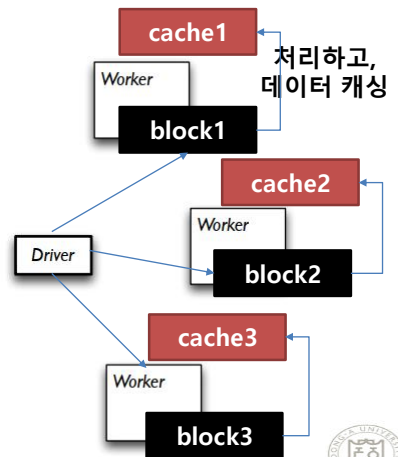
```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.filter(_.contains("mysql")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

// action 2
messages.filter(_.contains("php")).count()
```

실행 완료



Log Mining Example

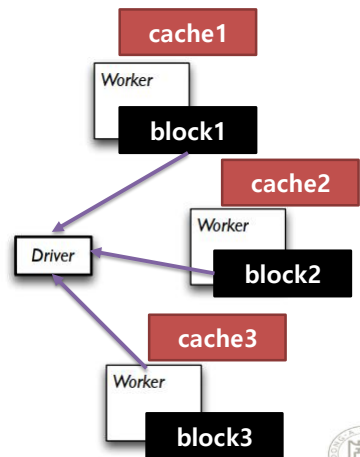
```
// base RDD
val lines = sc.textFile("hdfs://...")

// transformed RDDs
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.filter(_.contains("mysql")).map(r => r(1))
messages.cache()

// action 1
messages.filter(_.contains("mysql")).count()

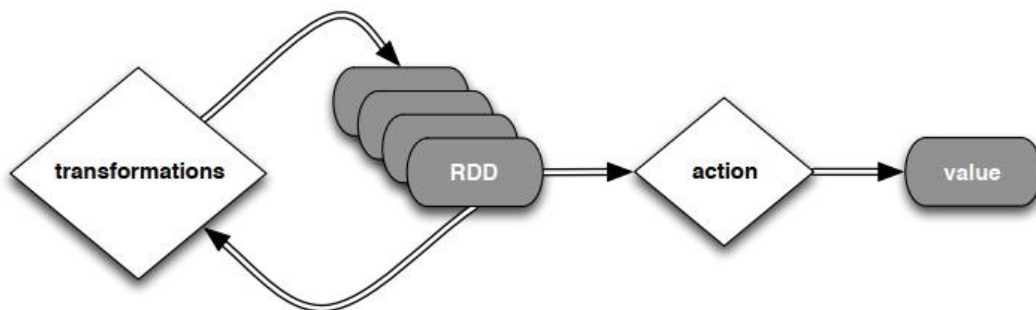
// action 2
messages.filter(_.contains("php")).count()
```

실행 완료



Spark Deconstructed

■ RDD transformations and actions



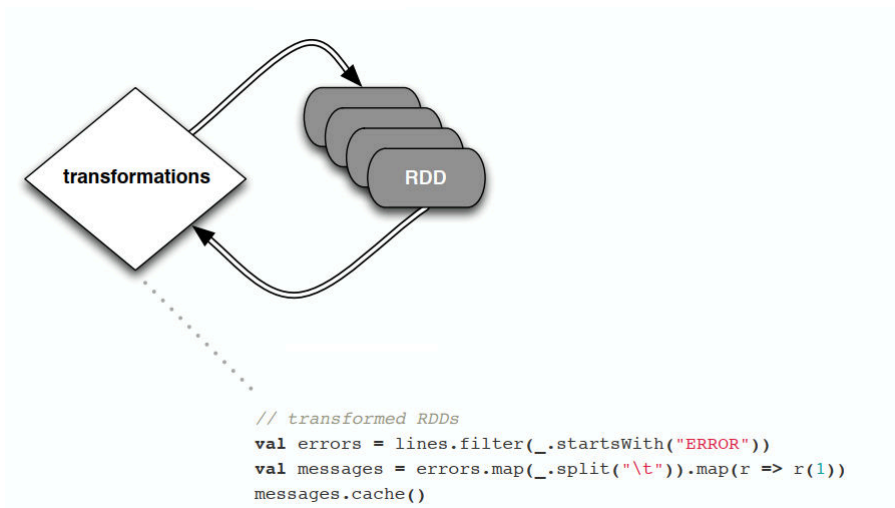
Spark Deconstructed



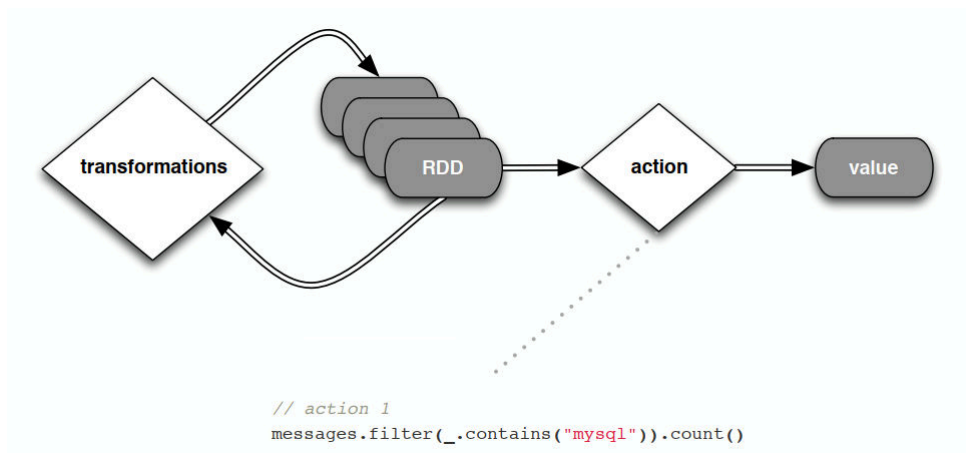
```
// base RDD
val lines = sc.textFile("hdfs://...")
```



Spark Deconstructed



Spark Deconstructed





Simple Spark Apps

01 시작하기

Word Count

- Text 문서의 콜렉션 내에서, 각 단어들이 얼마나 나타나는지를 세기
- 병렬적으로 처리하는 방법

```
void map (String doc_id, String text):
    for each word w in segment(text):
        emit(w, "1");

void reduce (String word, Iterator group):
    int count = 0;

    for each pc in group:
        count += Int(pc);

    emit(word, String(count));
```



Word Count

Scala:

```
val f = sc.textFile("README.md")
val wc = f.flatMap(l => l.split(" ")).map(word => (word, 1)).reduceByKey(_ + _)
wc.saveAsTextFile("wc_out.txt")
```

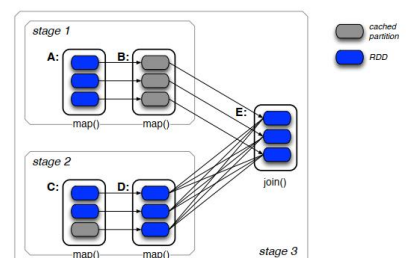
Python:

```
from operator import add
f = sc.textFile("README.md")
wc = f.flatMap(lambda x: x.split(' ')).map(lambda x: (x, 1)).reduceByKey(add)
wc.saveAsTextFile("wc_out.txt")
```



Word Count

2014-03-04	15dfb8e6cc4111e3a5bb600308919594	11		
2014-03-06	81da510acc4111e387f3600308919594	61		
2014-03-02	15dfb8e6cc4111e3a5bb600308919594	1	33.6599436237	-117.958125229
2014-03-04	81da510acc4111e387f3600308919594	2	33.8570099635	-117.855744398



Source Code

```
val format = new java.text.SimpleDateFormat("yyyy-MM-dd")

case class Register (d: java.util.Date, uuid: String, cust_id: String, lat: Float, lng: Float)
case class Click (d: java.util.Date, uuid: String, landing_page: Int)

val reg = sc.textFile("reg.tsv").map(_._split("\t")).map(
  r => (r(1), Register(format.parse(r(0)), r(1), r(2), r(3).toFloat, r(4).toFloat))
)

val clk = sc.textFile("clk.tsv").map(_._split("\t")).map(
  c => (c(1), Click(format.parse(c(0)), c(1), c(2).trim.toInt))
)

reg.join(clk).take(2)
```



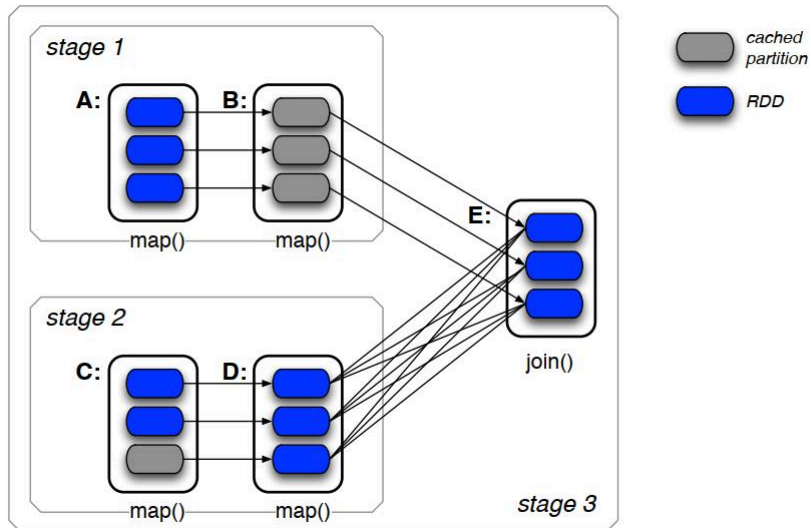
Source Code

■ Operator graph 생성

```
scala> reg.join(clk).toDebugString
res5: String =
FlatMappedValuesRDD[46] at join at <console>:23 (1 partitions)
  MappedValuesRDD[45] at join at <console>:23 (1 partitions)
    CoGroupedRDD[44] at join at <console>:23 (1 partitions)
      MappedRDD[36] at map at <console>:16 (1 partitions)
        MappedRDD[35] at map at <console>:16 (1 partitions)
          MappedRDD[34] at textFile at <console>:16 (1 partitions)
            HadoopRDD[33] at textFile at <console>:16 (1 partitions)
          MappedRDD[40] at map at <console>:16 (1 partitions)
            MappedRDD[39] at map at <console>:16 (1 partitions)
              MappedRDD[38] at textFile at <console>:16 (1 partitions)
                HadoopRDD[37] at textFile at <console>:16 (1 partitions)
```



Operator Graph



실습

■ Github의 README.md와 CHANGES.txt를 사용

- 특정 키워드를 가진 라인에 대해서 FILTER하는 RDD 생성
- 각 라인에 대해 Word Count를 수행
- 두 RDD간 조인(Join)

